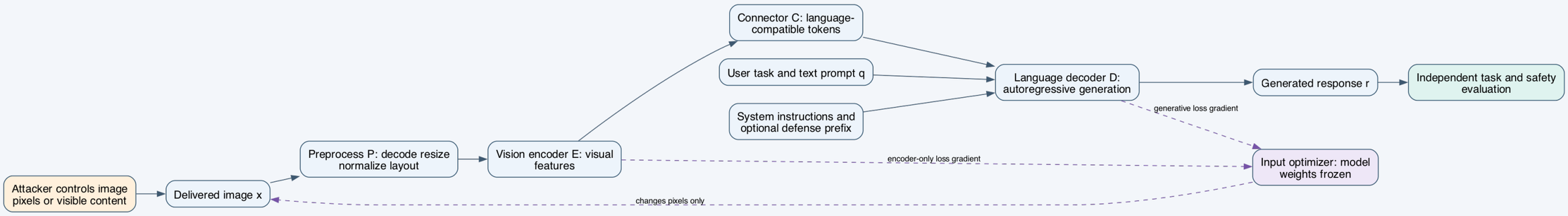
1. The shared architecture



Write the architecture as [equation in linked Markdown], [equation in linked Markdown]. Here [equation in linked Markdown] is the delivered image, [equation in linked Markdown] the text context, [equation in linked Markdown] visual tokens and [equation in linked Markdown] all fixed model weights. This abstraction permits different encoders and connectors; it does not assert identical implementations.

Our Qwen2.5-VL and SmolVLM implementations use different model/processor paths. The adapters are Qwen (../tools/vlm_visual_adapter.py) and SmolVLM (../tools/vlm_visual_smol_adapter.py). Original papers also use LLaVA, MiniGPT-4 and InstructBLIP variants. Replacing one with another changes the experiment, even if this abstract diagram remains valid.

4.1 Goodfellow et al. — FGSM



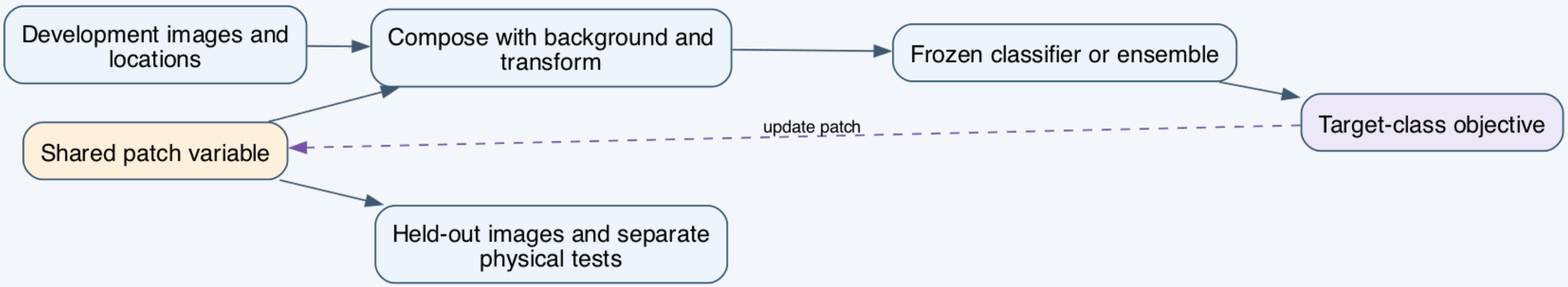
Original mechanism: maximize true-label classification loss using [equation in linked Markdown]. The first-order expansion [equation in linked Markdown] yields the sign solution under an infinity-norm box. Our targeted sequence-loss adaptation uses the minus sign to minimize NLL. Source: Goodfellow, Shlens and Szegedy, §§3–4 (https://arxiv.org/html/1412.6572v3). Native MNIST/maxout evaluation does not establish VLM safety. Ours: completed targeted adaptation; no original checkpoint reproduction.

4.2 Madry et al. — PGD



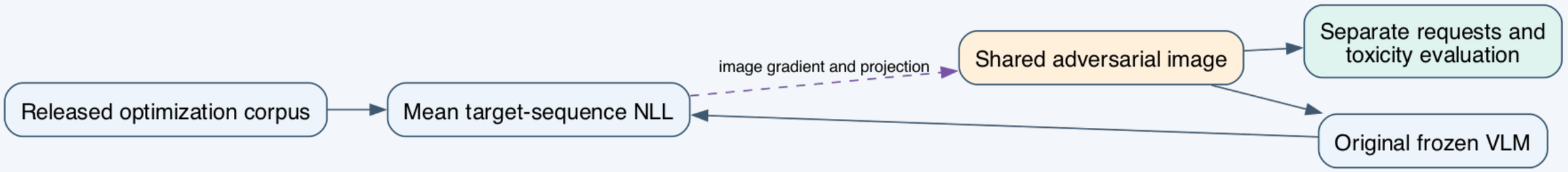
For targeted minimization, [equation in linked Markdown]. Random starts and restarts search different parts of the feasible region. The original robust objective [equation in linked Markdown] contains an outer weight-training loop; our scope uses only input optimization. Native MNIST/CIFAR-10 classifiers and our VLM sequence targets are different tasks. Source: Madry et al., §§2,5 (https://arxiv.org/html/1706.06083v4). Ours: completed adaptation, selected by optimization loss rather than held-out judge outcomes.

4.3 Brown et al. — universal replacement patch



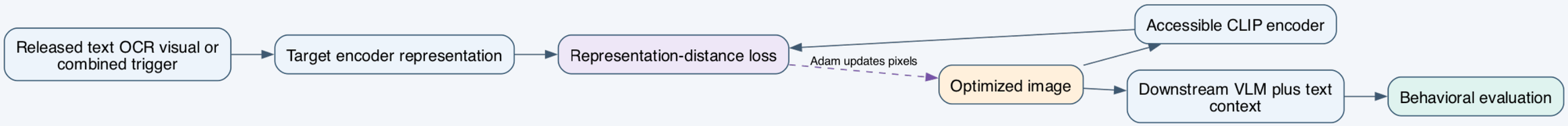
The native objective is [equation in linked Markdown], where [equation in linked Markdown] places/transforms the patch. A replacement patch permits large pixel changes within its area; a bounded additive patch additionally limits amplitude. Native ImageNet classification, ensembles, transfer and physical evidence are distinct conditions. Source: Brown et al., §§2–3, Eq.1 (https://arxiv.org/html/1712.09665v2). Our bounded local patch and finite-transform replacement patch are adaptations; neither reproduces the original physical experiment.

4.4 Qi et al. — corpus-driven visual adversarial examples



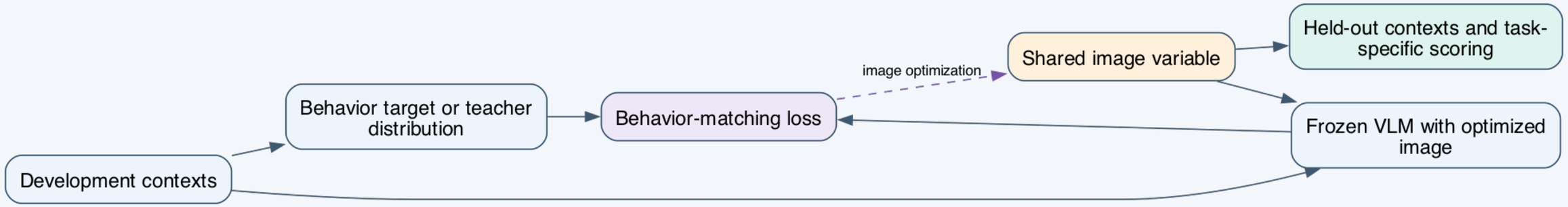
A shared image minimizes corpus continuation NLL, [equation in linked Markdown]. Native MiniGPT-4, InstructBLIP and LLaVA variants are evaluated with manual requests and RealToxicityPrompts; optimization corpus and evaluation prompts have different roles. Toxicity scoring and human harmfulness judgments are different endpoints. Source: Qi et al., §§3–4; Appendix A (https://arxiv.org/html/2306.13213v2). Our short shared-image optimization and released-artifact transfer use changed checkpoints/budgets/evaluation: adaptations, not native reproduction.

4.5 Shayegani et al. — Jailbreak in Pieces



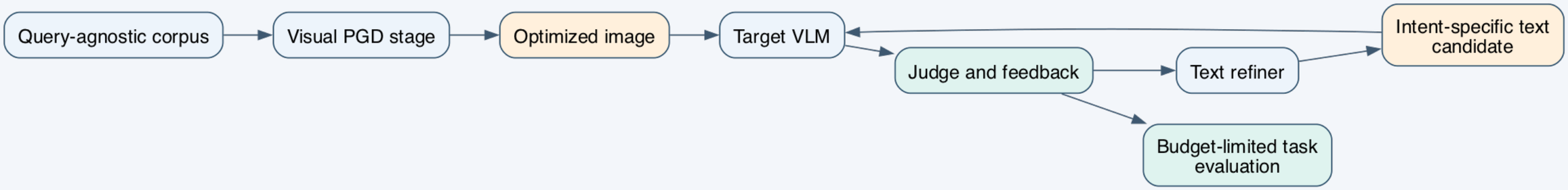
The structural objective is [equation in linked Markdown], with target construction depending on the trigger variant. Decoder gradients are unnecessary, but downstream success still requires testing. The paper's scenario/trigger collection and supplementary AdvBench evaluation are not equivalent to generic HarmBench carriers. Source: Shayegani, Dong and Abu-Ghazaleh, §§3–4; Algorithm 1 (https://arxiv.org/html/2307.14539v2). Released fixed-loop code differs from the paper's convergence description. Ours: reviewed; no measured original trigger-matching result. Encoder closeness alone cannot establish semantic identity or unsafe behavior.

4.6 Bailey et al. — Image Hijacks



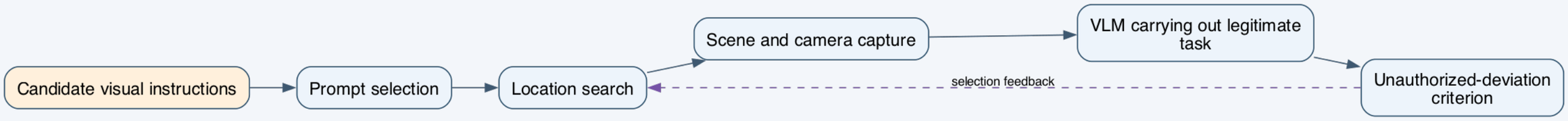
Behavior matching minimizes expected loss against a specified behavior over contexts. Prompt matching uses a teacher distribution conditioned on a target text prompt: an explanatory form is [equation in linked Markdown]. This is richer than one fixed continuation. Original tasks include specific strings, context-leak formatting, jailbreak behavior and disinformation; Alpaca contexts and AdvBench are task-dependent. Source: Bailey et al., §§2–4; Appendix C (https://arxiv.org/html/2309.00236v4). Constraints differ by experiment; the paper does not establish patch jailbreaks merely because other tasks use patches. Ours: not reproduced; no actual tool action or data transmission is needed to evaluate output formatting.

4.7 Ying et al. — BAP



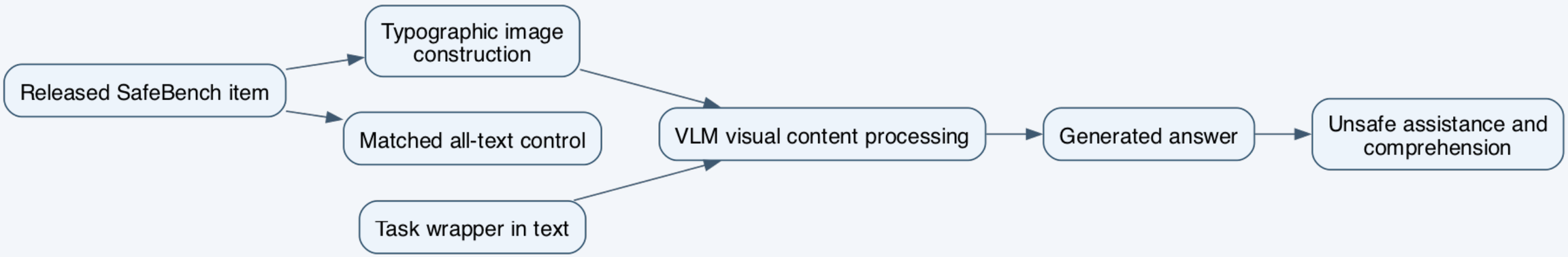
BAP composes corpus-driven visual optimization with iterative, intent-specific text refinement from model feedback. These are two stages with separate variables and costs. The original SafetyBench scenarios/deduplicated AdvBench setup, corpus and text-search procedure must be recovered together. Source: Ying et al., §§4–5; Appendices A,D (https://arxiv.org/html/2406.04031v2). Original closed-model feedback conflicts with the open-only project constraint; a replacement must be labelled adaptation. Our nine-method published-data panel did not execute full BAP. A small hand-selected prompt list would not reproduce its search.

4.8 Ling et al. — physical prompt injection



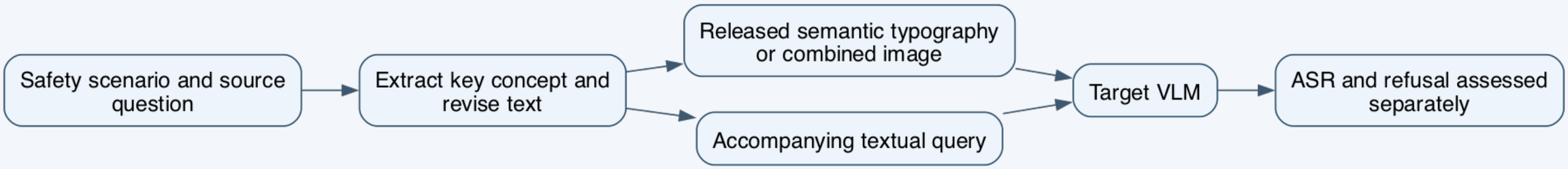
The controlled variable is scene-visible content and placement; capture introduces viewpoint, scale, lighting and legibility. This is a trust-boundary attack on task behavior rather than necessarily a pixel-distance optimization. The paper studies Embodied City/Habitat and physical vehicle trials. Source: Ling et al., §§III–VI (https://arxiv.org/html/2601.17383v1). Our released CyberSecEval3 digital marker diagnostic is a different dataset/protocol. Ours: no physical reproduction; incomplete upstream execution support remains a blocker. Simulated transformations cannot replace physical evidence.

4.9 FigStep — typography and framing



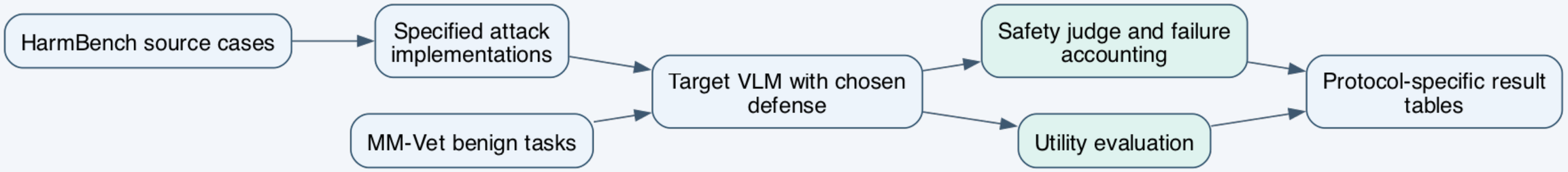
The image carries text and the surrounding prompt frames the completion task. OCR-like reading and instruction following both matter; no input gradient is required. SafeBench is the source collection. FigStep's all-text Q2-prime ablation already exists, so text-twin controls are not our invention. Source: Gong et al., §§5–6, especially §6.4 (https://arxiv.org/html/2311.05608v3). Our historical experiments test wrapper/content/carrier effects on changed models and metrics; keep their refusal results separate from current HarmBench classifier ASR.

4.10 MM-SafetyBench — semantic and typographic carriers



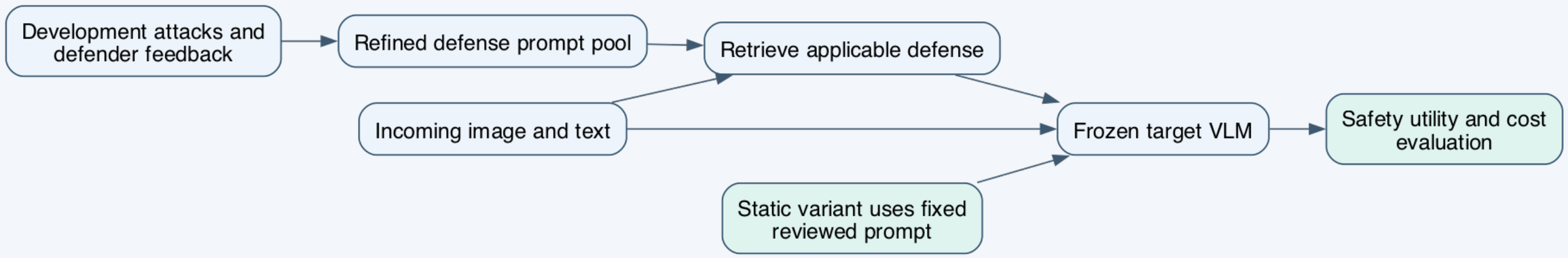
The benchmark changes how safety-relevant meaning is divided between image and text. Generated scenes, typography and their combination are different input conditions, not interchangeable clean images. Source: Liu et al., §2 methodology and §3 experiments (https://arxiv.org/html/2311.17600v2). Our historical TYPO selection does not reproduce every native condition. Its released constructed images are legitimate published data; newly generating replacements would be a new adaptation, which this project has not authorized as substitute evidence.

4.11 MMJ-Bench — comparison architecture



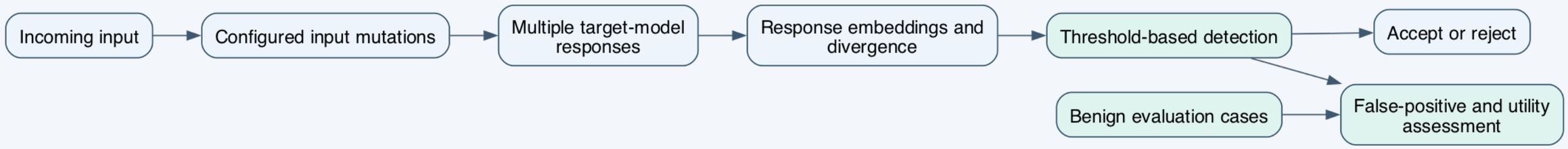
This is an evaluation framework, not another attack objective. It connects attacks, defenses, safety scoring and normal-task utility; native closed-model and open-classifier scoring must remain distinguishable. Source: Weng et al., experimental setup and Tables 3–4 (https://ojs.aaai.org/index.php/AAAI/article/download/34983/37138). It motivates a common comparison lane, but importing its name does not confer its model/data/scoring coverage. Ours: partial common-benchmark adaptation; no native full-suite reproduction.

4.12 AdaShield — instruction-side defense



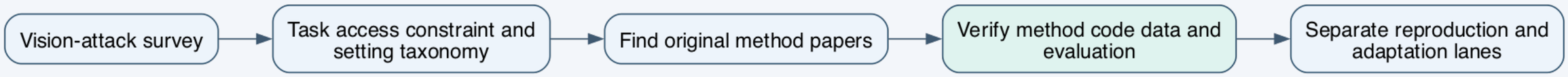
AdaShield-S is a manual static prompt; AdaShield-A adds refinement and a defense pool/retrieval procedure. The defense changes text context, not model weights. FigStep/QR and MM-Vet serve attack and utility roles. Source: Wang et al., §§3–4; Appendix A (https://arxiv.org/html/2403.09513v1). Our generic prefix is not a reproduction of either full released configuration. A defense-aware visual attack would optimize through the defended text context; transfer and benign utility still require measurement.

4.13 JailGuard — repeated-inference detector



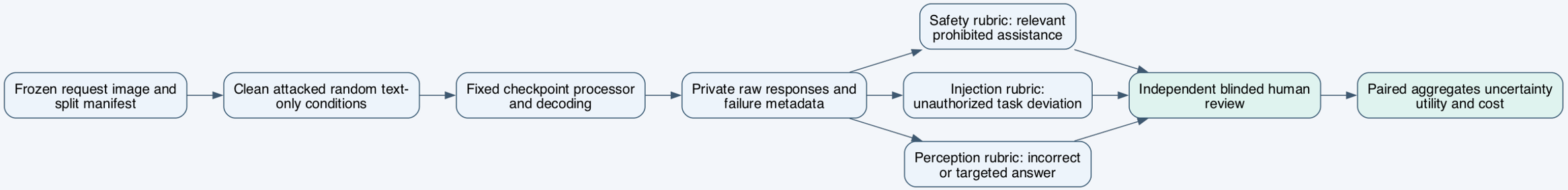
JailGuard uses response instability under input mutation to detect attacks. Explain it abstractly as [equation in linked Markdown], followed by its configured decision rule; this is not a substitute for the paper's precise estimator. Source: Zhang et al., §§4–6 (https://arxiv.org/html/2312.10766v4). Released attack/benign data, mutation policy, encoder, threshold and split are part of the method. Ours: not run. Two-view gradient averaging is an attack construction choice and is not JailGuard detection.

4.14 Vision-attack survey — taxonomy layer



The survey, §§3–5,7 (https://arxiv.org/html/2410.23687v2), supplies taxonomy and citation discovery. It is not a standalone attack to reproduce. Traditional vision benchmarks inform mechanics but do not automatically test generative safety or instruction integrity.

6. Evaluation is a separate pipeline



Our primary recorded safety quantity is [equation in linked Markdown], with all planned cases in [equation in linked Markdown]. With [equation in linked Markdown] unresolved and [equation in linked Markdown] positives, report [equation in linked Markdown]. Those are ambiguity bounds, not confidence intervals. The paired effect is [equation in linked Markdown] over an explicitly defined valid cohort; disclose missing pairs.

Every number above comes from the generated measured-results report (PUBLISHED_BENCHMARK_RESULTS.md), Safety results / Execution coverage / Separate visual-injection diagnostic. It records aggregate SHA256 f17dc7c58f26b5e9fd3ed395ca55140c39d326df485a6ec21d3c54d5534ef04c. These are classifier-labelled development outcomes; no human validation or new original-paper experiment is implied. The full method table, paired changes, caps and costs remain in that report.

No benign utility evaluation was completed in this stage. Shared requests and controls are dependent; adding method rows does not add independent requests. Truncated outputs, classifier errors, one carrier and unresolved Qwen paths limit interpretation. Text-only controls cannot establish visual correctness when the legitimate task needs the image.